1. Competency #9: Understand the relationship between the context of the problem and the representation of the mathematical solution; use this understanding to interpret the solution in context and assess feasibility/limitations

> *"Understand the relationship between the context... and representation... interpret the solution in context..."*

- Highly relevant: The core of the question lies in interpreting a visual data representation (bar chart) within a real-world scenario (music sales over time).
- Students must map bars to specific bands and months, then compare their heights—not just read values, but relate them meaningfully (i.e., "more than" implies inequality comparison across categories over time).
- The phrase "for the first time" adds a temporal logic layer, requiring sequential scanning from January onward—this is only meaningful if the student understands the real-world timeline embedded in the graph.

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2. Competency #1: Interpret a mathematical result back into the real-world context

> *"Interpret a mathematical result back into the real world context"*

- Highly relevant: The answer (e.g., "April") is not just a label on an axis—it represents a real calendar month in which a market event occurred (one band outsold another for the first time).
- Correct interpretation requires recognizing that the mathematical outcome ("April is the first month where bar_NOD > bar_TKK") corresponds to a meaningful business or cultural milestone.

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3. Competency #5: Interpret mathematical outcomes in various formats; compare or evaluate multiple representations in relation to a situation

> *"Interpret... compare or evaluate two or more representations in relation to a situation"*

- Strongly relevant: Although only one visual format (bar chart) is used, the task inherently involves comparing two data sets (No One's Darling vs. The Kicking Kangaroos) across multiple time points.
- This is a form of multi-representational reasoning within a single graph: each month contains paired bars, and the student must evaluate relative magnitudes repeatedly.
- The skill aligns with evaluating "two or more representations" in the sense of two evolving quantitative narratives presented side by side.

4. Competency #2: Identify whether a mathematical conclusion makes sense given the problem context

> *"Identify whether a mathematical result... does, or does not, make sense given the context"*

- Moderately relevant: After identifying, say, April as the answer, a thoughtful student might check:
  - Was No One's Darling even active in January? (No—their CD released in February.)
  - Did The Kicking Kangaroos have high initial sales that gradually declined while No One's Darling grew?
- This contextual plausibility check—ensuring the crossover aligns with release dates and trends—engages Competency #2.
- The question design implicitly rewards attention to the narrative context (release timing), making reasonableness checks valuable.

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5. Competency #7: Construct and communicate explanations and arguments in the context of the problem

> *"Construct and communicate explanations and arguments in the context of the problem"*

- Mildly relevant: While the question is multiple-choice and doesn't require a written justification, the cognitive process involves constructing an implicit argument:
  "In Jan/Feb, TKK > NOD; in March, still TKK ≥ NOD; in April, NOD > TKK → so April is the first time."
- In an open-response setting, this would directly invoke Competency #7. Even in MCQ form, the reasoning structure mirrors argument construction.

1. 第9项能力：理解问题情境与数学表示之间的关系，并据此解释解的现实含义及可行性/局限性

> “理解问题情境与数学解表示之间的关系……据此解释解在具体情境中的意义。”

- 高度相关：本题的核心在于将可视化数据（柱状图）置于真实世界背景（音乐销量随时间变化）中进行解读。
- 学生不仅要识别各柱子对应的乐队和月份，还需进行有意义的比较（即“销量更多”意味着数值不等式的判断），而不仅仅是读取单个数值。
- “首次”这一关键词引入了时间逻辑——必须从1月开始逐月检查，直到找到第一个满足条件的月份。这种顺序扫描只有在理解图表所承载的真实时间线时才有意义。

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2. 第1项能力：将数学结果还原到现实世界的情境中

> “将数学结果重新解释回现实世界的情境中。”

- 高度相关：答案（例如“4月”）不仅仅是一个坐标轴上的标签，而是代表现实中某个月份发生的一个市场事件（一支乐队首次销量超过另一支）。
- 正确作答需要认识到：数学结论（“4月是 No One’s Darling 首次销量高于 The Kicking Kangaroos 的月份”）对应着一个具有实际意义的文化或商业节点。

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3. 第5项能力：以多种格式解读数学结果，并根据情境比较或评估不同表示

> “以多种格式解读数学结果……根据情境比较或评估两个及以上表示。”

- 强相关：尽管题目只使用了一种视觉形式（柱状图），但任务本质上涉及对两条数据序列的持续比较（No One’s Darling vs. The Kicking Kangaroos）。
- 这是一种在同一图表内进行多表示推理的过程：每个月都包含一对柱子，学生需反复判断其相对高度。
- 虽然未切换图表类型，但“两条随时间变化的数据轨迹”可视为两种动态表示，符合“比较多个表示”的精神。

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4. 第2项能力：判断数学结论在给定情境下是否合理

> “判断数学结果或结论在特定情境中是否合理。”

- 中等相关：在得出“4月”为答案后，有思考习惯的学生可能会验证：
  - No One’s Darling 在1月尚未发行CD（2月才发行），因此1月不可能超越；
  - The Kicking Kangaroos 作为1月发行的乐队，初期销量可能较高，随后下降，而 No One’s Darling 可能逐步上升；
- 这种基于发行时间与趋势的合理性检验，正是第2项能力的体现。题目虽未明示，但情境信息（发行月份）为判断提供了依据，增强了答案的可信度。

5. 第7项能力：在问题情境中构建并交流解释或论证

> “在问题情境中构建并交流解释或论证。”

- 弱相关但存在：虽然题目是选择题，不要求书面解释，但正确作答的思维过程隐含了一个逻辑论证结构：

  > “1月、2月：TKK > NOD（或NOD未发行）；3月：TKK ≥ NOD；4月：NOD > TKK → 因此4月是首次。”

- 若题目改为开放题（如“请说明你的推理”），则此项能力会成为核心。即便在选择题中，其认知基础仍依赖于论证构建。